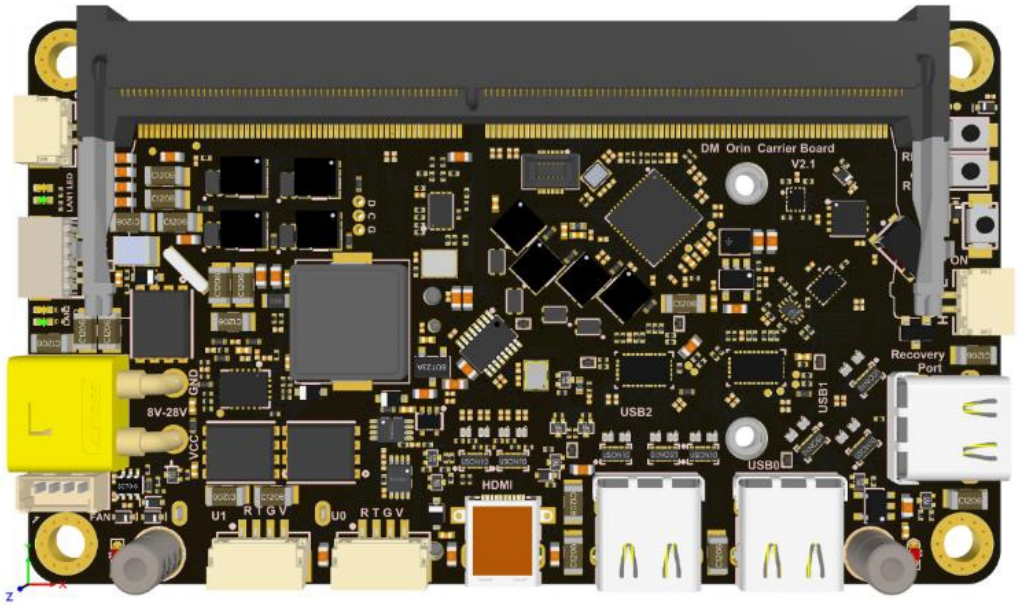
DAMIAO|达妙科技

DM-ORIN NX V2.X

Carrier Board

User Manual V1.0 2025-03-03



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Table of Contents

Revision History 3
Disclaimer 4
Precautions 4
Carrier Board Overview 4
Specifications 5
Interface Description 6
◈ Power Tree 6
◈ UART 7
◈ CAN 7
◈ Fan Power Connector 8
◈ Controlled Fan Power Connector 8
◈ USB2.0（SH1.0-5pin） 8
◈ USB3.0-0（TypeC） 9
◈ USB3.0-1（TypeC） 9
◈ USB3.0-2（TypeC） 10
◈ Key 10
◈ Gigabit Ethernet Port 10
◈ CIS0 12
◈ CIS1 13
◈ LED 14
◈ GPIO 14
Appendix I. CAN Interface Usage 15
Appendix II: RTL8125 Driver Installation 17
Appendix III: Dimensional Drawing 18

Revision History

Date	Version	Changes
2025-03-03	V1.0	Initial Release

Disclaimer

Thank you for choosing the DAMIAO DM-ORIN NX V2.X Carrier Board (The"X"represents a digit from 0 to 9.) from DAMIAO Technology.This product is hereinafter referred to as the "Carrier Board".

Before using this product, please read this manual and all safety precautions provided by DAMIAO Technology and follow them strictly. Failure to do so may result in personal injury, product damage, or damage to surrounding property. By using this product, you acknowledge that you have read, understood, and agreed to all terms and conditions in this document and all related documentation. You agree to use this product only for lawful and intended purposes. You assume full responsibility for the use of this product and any consequences arising from it. DAMIAO Technology shall not be liable for any direct or indirect damage, injury, or legal liability arising from the use of this product.

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Precautions

1. Connect all cables according to the instructions in this manual before use to prevent damage to the connectors or the carrier board.

2. Operate the carrier board within the recommended operating conditions, including voltage, current, and temperature limits, to prevent damage and ensure product service life.

3. Inspect all components before use.Do not operate the motor if any components are missing, worn, or damaged.

4. Take appropriate protective measures during operation. Do not touch components on the carrier board directly with your hands. Protect against electrostatic discharge and mechanical damage. Keep the carrier board clean to prevent short circuits or performance degradation.

5. In case of abnormal conditions such as sparking, smoke, or burning smell during operation, immediately disconnect the power supply and stop using the product.

Carrier Board Overview

The DM-ORIN NX V2.X Carrier Board is a compact carrier board developed by DAMIAO Technology for lightweight equipment. It is compatible with the NVIDIA® JETSON ORIN NX ™ series modules, weighs only 39.2 g, supports a wide input voltage range, and supports ORIN SUPER mode.

With a wide range of interfaces, the carrier board is suitable for applications with strict size requirements, such as UAVs, handheld surveying and mapping devices, and legged robots.

Features:

- High compatibility
- Supports Jetson Orin NX and Jetson Orin Nano modules
- Supports ORIN SUPER mode

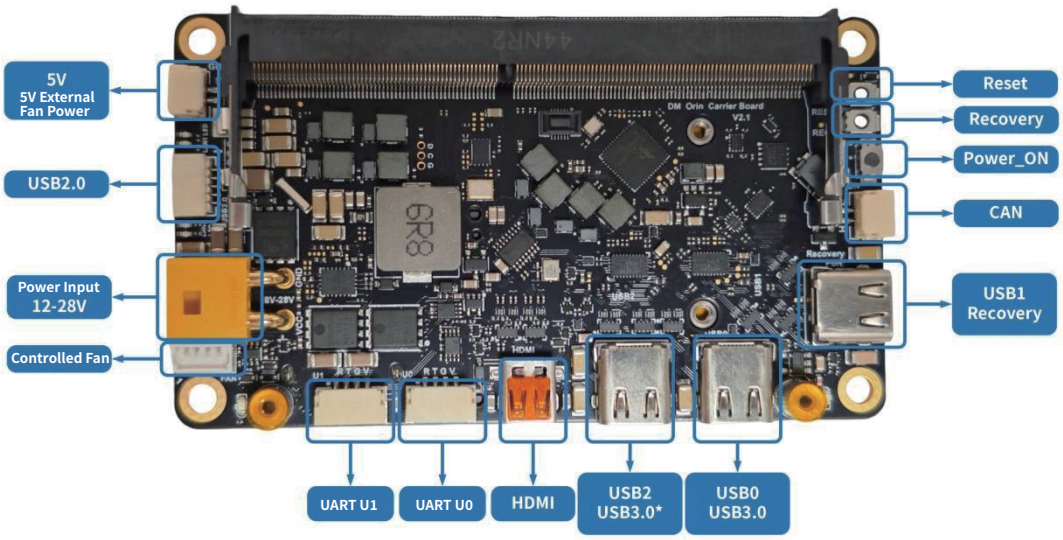
Note: The Jetson Xavier NX supports only one USB 3.0 interface and does not support the extended Ethernet interface.

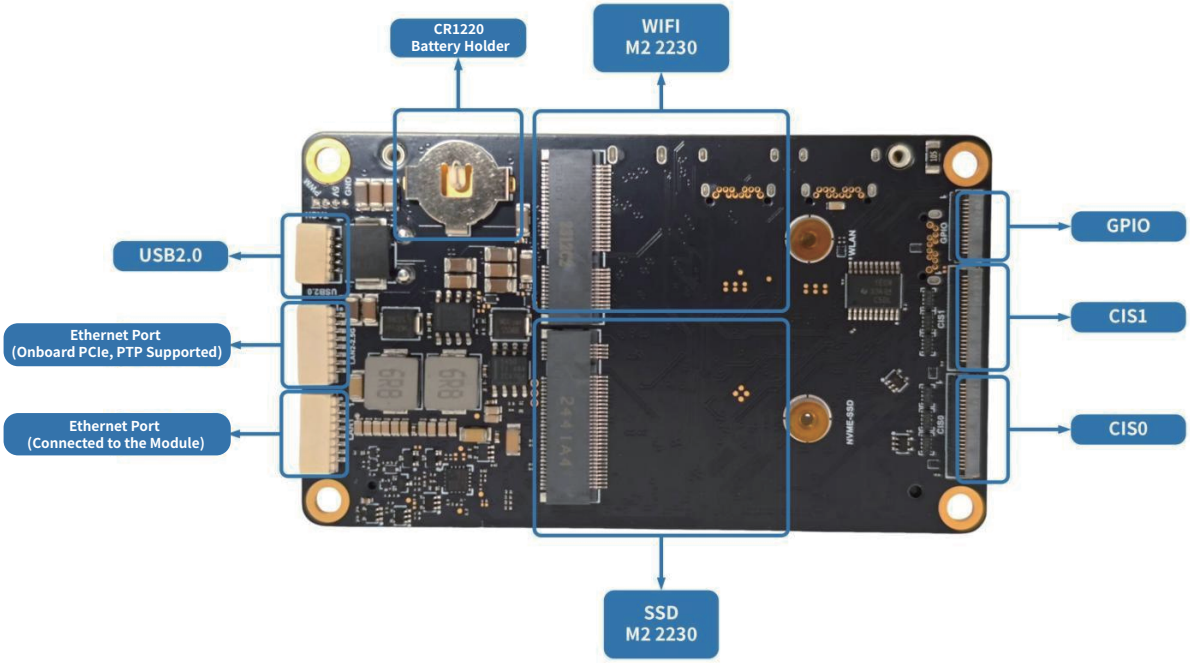
Specifications

No.	Item	Specification	Remarks
1	Weight	39.2g	–
2	Input Voltage	12~28 V	Support 6S Lipo battery input
3	Dimensions	50x86.8x13mm	–
4	Operating Temperature	-20° C to 50° C	–

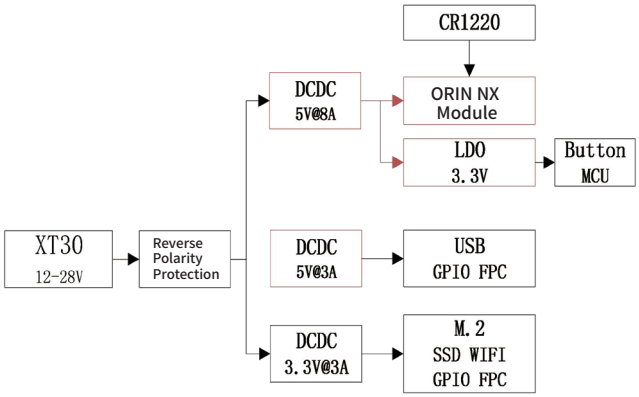
No.	Interface	Type	Qty.	Remarks
1	USB3.0	TypeC	3	One supports Recovery
2	USB2.0	SH1.0(5pin)	2	–
3	HDMI	MicroHDMI	1	–
4	Ethernet	SH1.0(8pin)	1	Onboard transformer
		SH1.0(8pin)	1	Onboard transformer, up to 2.5 Gbps, PTP supported
5	M.2	M.2 KeyM	1	2230 SSD
		M.2 KeyE	1	2230 Wi-Fi card（AW-CB375NF)
6	CAN	GH1.25 2pin	1	Up to 5 Mbps; no termination resistor
7	UART	GH1.25 4pin	2	3.3 V logic level; 5 V power output
8	CIS	0.5mmFPC 22Pin	2	same pinout as the original design,supports top- and bottom-side insertion
9	GPIO	0.5mmFPC 14Pin	1	SPI1、IIC1*、GPIO07 3.3 V logic level
10	Fan	1.25 4pin	1	Same as the original version
11		GH1.25 2pin	1	5 V power output only
12	Push Button	Tactile switch	3	ReSet、Recovery、PowerON
13	RTC BAT	MY1220	1	CR1220 coin cell

Interface Description





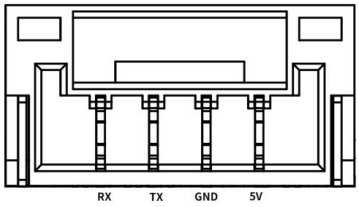
◈ Power Tree



◈ UART

UART（UART0）			
Pin	Signal	Module Pin	Type
1	RX	101	3.3 V Logic
2	TX	99	3.3 V Logic
3	GND	--	Power
4	5V	--	Power

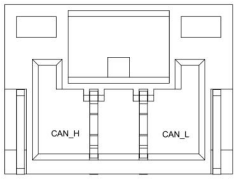
UART（UART1）			
Pin	Signal	Module Pin	Type
1	RX	205	3.3 V Logic
2	TX	203	3.3 V Logic
3	GND	--	Power
4	5V	--	Power



(Figure 1—"UART" Interface Diagram)

◈ CAN

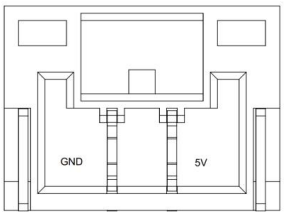
CAN			
Pin	Signal	Module Pin	Type
1	CAN_H	--	Differential
2	CAN_L	--	Differential
Note: The CAN transceiver is SIT1042 and supports a maximum baud rate of 5 Mbps.			



(Figure 2—"CAN" Interface Diagram)

◈ Fan Power Connector

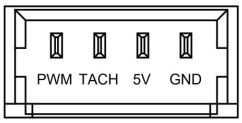
Fan Power Connector（same supply as the module）			
Pin	Signal	Module Pin	Type
1	GND	--	Power
2	5V	--	Power



(Figure 3—"Fan Power Connector" Interface Diagram)

◈ Controlled Fan Power Connector

Controlled Fan Power Connector（same supply as the module）			
Pin	Signal	Module Pin	Type
1	PWM	230	5 V Logic
2	TACH	208	5 V Logic
3	5V	--	Power
4	GND	--	Power

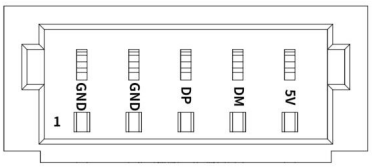


(Figure 4—"Controlled Fan Power Connector" Interface Diagram)

◈ USB2.0（SH1.0-5pin）

USB2.0（SH1.0-5pin）			
Pin	Signal	Module Pin	Type
1	GND	--	Power
2	GND	--	Power
3	USB-DP	--	USB Differential
4	USB-DM	--	USB Differential
5	5V	--	Power

USB2.0（SH1.0-5pin）
Note: The USB 2.0 interface is split from the module's USB2 signals (pins 121 and 123) through a USB hub.



(Figure 5—"USB2.0（SH1.0-5pin）" Interface Diagram)

◈ USB3.0-0（TypeC）

USB3.0-0（TypeC）			
Pin	Signal	Module Pin	Type
1	USBSS_TX_P	168	USB 3.0 Differential
2	USBSS_TX_N	166	USB 3.0 Differential
3	USBSS_RX_P	163	USB 3.0 Differential
4	USBSS_RX_N	161	USB 3.0 Differential
5	USB1_DP	117	USB 2.0 Differential
6	USB1_DM	115	USB 2.0 Differential

◈ USB3.0-1（TypeC）

USB3.0-1（TypeC）			
Pin	Signal	Module Pin	Type
1	USBSS1_TX_P	47	USB 3.0 Differential
2	USBSS1_TX_N	45	USB 3.0 Differential
3	USBSS1_RX_P	41	USB 3.0 Differential
4	USBSS1_RX_N	39	USB 3.0 Differential
5	USB0_DP	111	USB 2.0 Differential
6	USB0_DM	109	USB 2.0 Differential
7	I2C0_SCL	185	Open-Drain
8	I2C0_SDA	187	Open-Drain
9	USB_VBUS_DET*	87	GPIO
Note: This USB interface is used as the Recovery interface.			

◈ USB3.0-2（TypeC）

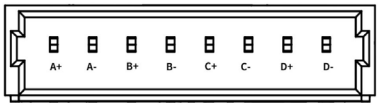
USB3.0-2（TypeC）			
Pin	Signal	Module Pin	Type
1	USBSS2_TX_P	59	USB 3.0 Differential
2	USBSS2_TX_N	57	USB 3.0 Differential
3	USBSS2_RX_P	53	USB 3.0 Differential
4	USBSS2_RX_N	51	USB 3.0 Differential
5	USB_DP	117	USB 2.0 Differential
6	USB_DM	115	USB 2.0 Differential
Note 1: The USB 2.0 interface is split from the module's USB2 signals (pins 121 and 123) through a USB hub.			
Note 2: This USB 3.0 interface must be enabled in the device tree.			

◈ Key

Key			
Pin	Signal	Module Pin	Type
1	RECOVERY	214	--
2	SYS_RESET	239	--
3	POWER_ON	240	--

◈ Gigabit Ethernet Port

Gigabit Ethernet Port			
Pin	Connector Pin	Signal Before Magnetics	Module Pin
1	A+	GBE_MDIO_P	186
2	A-	GBE_MDIO_N	184
3	B+	GBE_MDI1_P	192
4	B-	GBE_MDI1_N	190
5	C+	GBE_MDI2_P	198
6	C-	GBE_MDI2_N	196
7	D+	GBE_MDI3_P	204
8	D-	GBE_MDI3_N	202

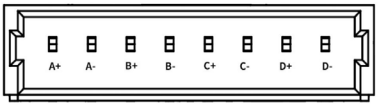


(Figure 6—"Gigabit Ethernet Port" Interface Diagram)

The Ethernet interface supports speeds of up to 2.5 Gbps and supports PTP.

PCIE Interface			
No.	Signal	Module Pin	Type
1	PCIE2_CLK_N	52	Differential
2	PCIE2_CLK_P	54	Differential
3	PCIE2_TX0_N	46	Differential
4	PCIE2_TX0_P	48	Differential
5	PCIE2_RX0_N	40	Differential
6	PCIE2_RX0_P	42	Differential
7	PCIE2_CLKREQ	211	Clock
8	PCIE2_RST	219	Reset

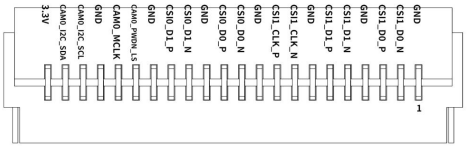
Pin	Connector	Type
1	A+	Differential
2	A-	Differential
3	B+	Differential
4	B-	Differential
5	C+	Differential
6	C-	Differential
7	D+	Differential
8	D-	Differential
Note: The Ethernet interface is implemented using an RTL8125 controller connected to the module via PCIE.		



(Figure 7—Interface Diagram)

◈ CIS0

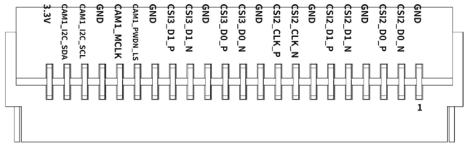
CIS0			
Pin	Signal	Module Pin	Type
1	GND	--	Power
2	CSI1_D0_N	3	CIS Differential
3	CSI1_D0_P	5	CIS Differential
4	GND	--	Power
5	CSI1_D1_N	15	CIS Differential
6	CSI1_D1_P	17	CIS Differential
7	GND	--	Power
8	CSI1_CLK_N	9	CIS Differential
9	CSI1_CLK_P	11	CIS Differential
10	GND	--	Power
11	CSI0_D0_N	4	CIS Differential
12	CSI0_D0_P	6	CIS Differential
13	GND	--	Power
14	CSI0_D1_N	16	CIS Differential
15	CSI0_D1_P	18	CIS Differential
16	GND	--	Power
17	CAM0_PWDN_LS	114	CMOS 1.8 V Output
18	CAM0_MCLK	116	CMOS 1.8 V Output
19	GND	--	Power
20	CAM0_I2C_SCL	*	Open-Drain, 3.3 V
21	CAM0_I2C_SDA	*	Open-Drain, 3.3 V
22	3.3V	--	Power
Note: This interface uses the same circuit design as the original. The IIC signals are switched from CAM_I2C_SDA (215) and CAM_I2C_SCL (213) through CAM_MUX_SEL (130).			



(Figure 8—"CIS0" Interface Diagram)

◈ CIS1

CIS1			
Pin	Signal	Module Pin	Type
1	GND	--	Power
2	CSI2_D0_N	22	CIS Differential
3	CSI2_D0_P	24	CIS Differential
4	GND	--	Power
5	CSI2_D1_N	34	CIS Differential
6	CSI2_D1_P	36	CIS Differential
7	GND	--	Power
8	CSI2_CLK_N	28	CIS Differential
9	CSI2_CLK_P	30	CIS Differential
10	GND	--	Power
11	CSI3_D0_N	21	CIS Differential
12	CSI3_D0_P	23	CIS Differential
13	GND	--	Power
14	CSI3_D1_N	33	CIS Differential
15	CSI3_D1_P	35	CIS Differential
16	GND	--	Power
17	CAM1_PWDN_LS	120	CMOS 1.8 V Output
18	CAM1_MCLK	122	CMOS 1.8 V Output
19	GND	--	Power
20	CAM1_I2C_SCL	*	Open-Drain, 3.3 V
21	CAM1_I2C_SDA	*	Open-Drain, 3.3 V
22	3.3V	--	Power
Note: This interface uses the same circuit design as the original. The IIC signals are switched from CAM_I2C_SDA (215) and CAM_I2C_SCL (213) through CAM_MUX_SEL (130).			



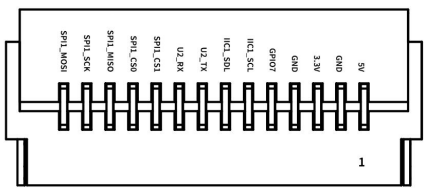
(Figure 9—"CIS1" Interface Diagram)

◈ LED

LED			
Pin	Signal	Module Pin	Type
1	PWR_LED_CTRL	127	--

◈ GPIO

GPIO			
Pin	Signal	Module Pin	Type
1	5V	--	Power
2	GND	--	Power
3	3.3V	--	Power
4	GND	--	Power
5	GPIO7	206	3.3 V
6	I2C1_SCL	189	3.3 V, Pull-Up
7	I2C1_SDA	191	3.3 V, Pull-Up
8	U2_TX	236	3.3 V
9	U2_RX	238	3.3 V
10	SPI1_CS1	97	3.3 V
11	SPI1_CS0	95	3.3 V
12	SPI1_MISO	93	3.3 V
13	SPI1_SCK	91	3.3 V
14	SPI1_MOSI	89	3.3 V



(Figure 10—"GPIO" Interface Diagram)

Appendix I. CAN Interface Usage

◈ M-Orin Carrier Board CAN Test

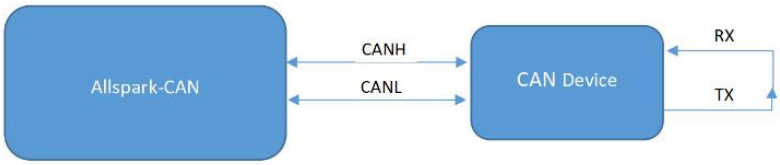
- Test Environment: **Ubuntu 20.04**
- Test Tools:

DM-Orin Carrier Board

DAMIAO USB-CAN Analyzer

- Procedures:

1. Connect the CAN InterfacesConnect the CAN port of the host computer to the custom CAN transceiver module. Connect CAN_H to CAN_H and CAN_L to CAN_L.



2. Activate CAN

Run the following command:

```
sudo modprobe mttcan
```

3. Configure the CAN Bit Rate

```
sudo ip link set can0 type can bitrate 1000000
```

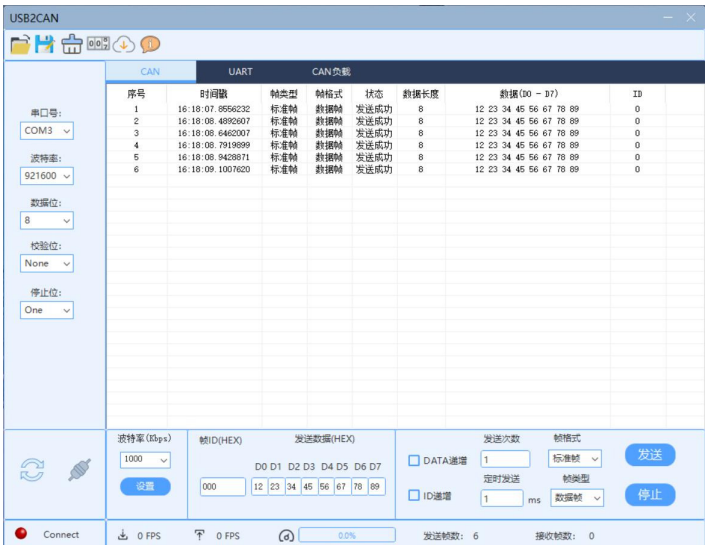
4. Bring Up the CAN Interface

```
sudo ip link set up can0
```

5. Receive CAN Frames

Monitor the can0 interface and display received CAN frames in the terminal.

```
sudo candump can0
```



```
nx@ubuntu:~/Desktop$ sudo modprobe mttcan
nx@ubuntu:~/Desktop$ sudo ip link set can0 type can bitrate 1000000
nx@ubuntu:~/Desktop$ sudo ip link set up can0
nx@ubuntu:~/Desktop$ sudo  candump can0
  can0  000   [8]  12 23 34 45 56 67 78 89
  can0  000   [8]  12 23 34 45 56 67 78 89
  can0  000   [8]  12 23 34 45 56 67 78 89
  can0  000   [8]  12 23 34 45 56 67 78 89
  can0  000   [8]  12 23 34 45 56 67 78 89
  can0  000   [8]  12 23 34 45 56 67 78 89
```

6. Send a CAN Frame

Use the following command to send a CAN frame with ID 001 and data 11223344AABBCCDD through the can0 interface.

```
cansend can0 001#11223344AABBCCDD
```

Appendix II: RTL8125 Driver Installation

Official Download Page:

https://www.realtek.com/Download/List?cate_id=584

Follow the installation instructions in the README file included in the downloaded archive.

tar vjxf r8125-9.aaa.bb.tar.bz2

cd r8125-9.aaa.bb

./autorun.sh (as root or with sudo)

Appendix III: Dimensional Drawing

